MILITARY SPECIFICATION SHEET

# RELAYS, HYBRID, TIME DELAY, VOLTAGE SENSITIVE, DELAY ON DROP-OUT, CLASS A, TYPE V, HERMETICALLY SEALED (POTTED), SPDT, 2 AMPERE, TRACK MOUNTED

This specification is approved for use within the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the relays described herein shall consist of this specification and the latest issue of MIL-R-83726.

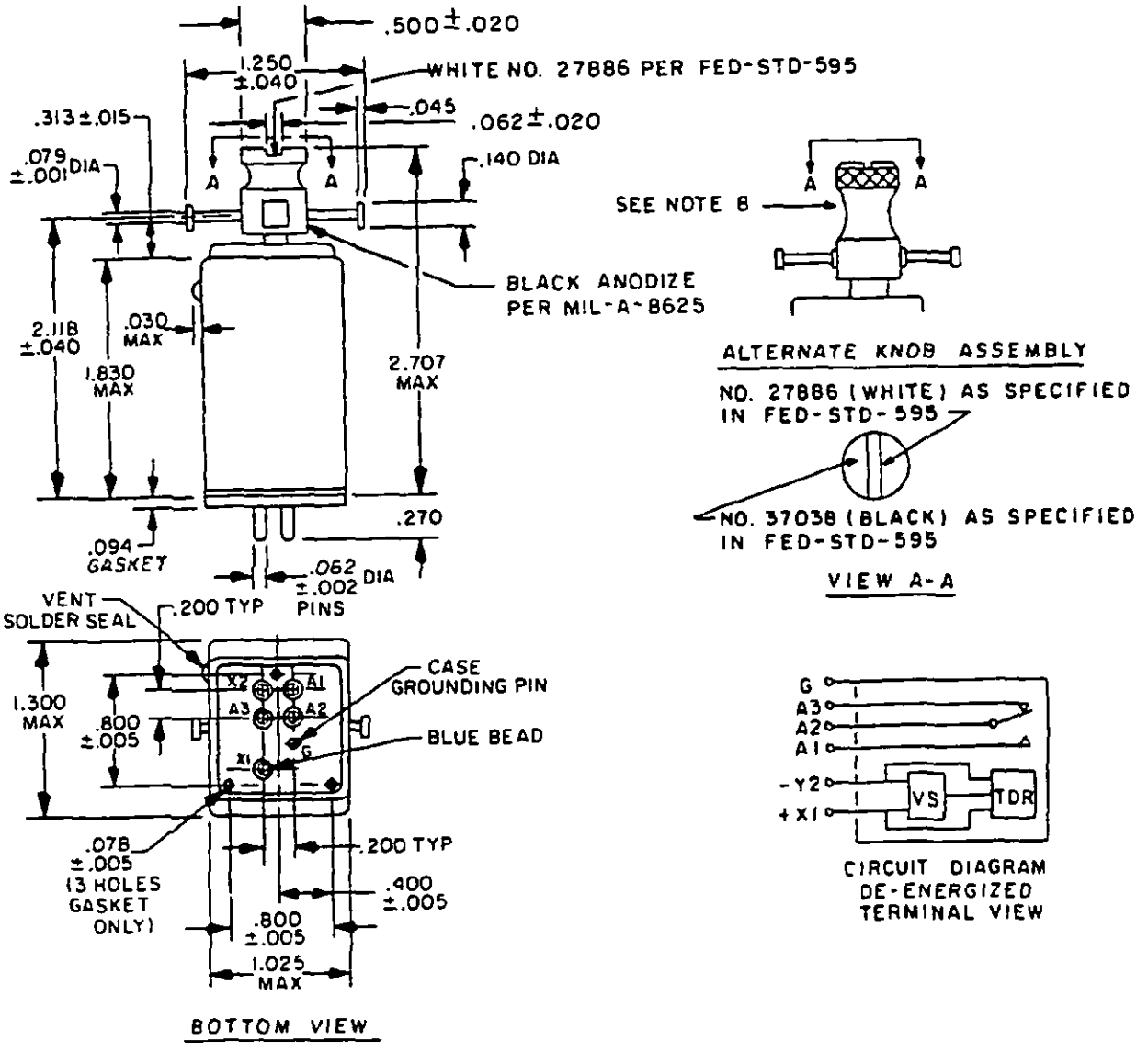



FIGURE 1. Outline dimensions and configuration of relay.

Ⓖ denotes changes


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| Inches | mm | Inches | mm |
|---|---|---|---|
| .001 | 0.03 | .140 | 3.56 |
| .002 | 0.05 | .200 | 5.08 |
| .005 | 0.13 | .270 | 6.86 |
| .015 | 0.25 | .313 | 7.95 |
| .020 | 0.51 | .400 | 10.16 |
| .030 | 0.76 | .500 | 12.70 |
| .040 | 1.02 | .800 | 20.32 |
| .045 | 1.14 | 1.025 | 26.04 |
| .062 | 1.57 | 1.250 | 31.75 |
| .078 | 1.98 | 1.300 | 33.02 |
| .079 | 2.01 | 1.830 | 46.40 |
| .094 | 2.39 | 2.118 | 53.80 |
| | | 2.707 | 68.76 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Terminal numbers shall not appear on the relay header and there shall be a legible circuit diagram on relay which identifies each terminal location specified.
4. For relay mounting track see MIL-R-6106/3 and for relay bracket and socket assembly see MIL-R-6106/5.
5. Gasket material: Methylvinyl silicone (red). Shore hardness A40.
6. Terminal plating - gold in accordance with MIL-G-45204 type II, class I over nickel plate.
7. Unless otherwise specified, tolerance is ±.010 (0.25 mm).
8. Knob and post shall be nickel plated (100 microinches thick minimum) 303 stainless steel.

FIGURE 1. Outline dimensions and configuration of relay - Continued.



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REQUIREMENTS:

Operating data:

Configuration: SPDT.

Life/load ratings: See table I.

Timing action: See table II.

TABLE I. Life/load ratings (relay case grounded).

| Type of load | Life (cycles) | Amperes 28 V dc |
|---|---|---|
| Resistive | 100,000 | 2.0 |
| Inductive | 50,000 | 0.5 |
| Lamp | 100,000 | 0.2 |
| Mechanical life (reduced current) | 400,000 | 0.5 |
| Intermediate current | 100,000 | 0.50 |

Contact drop:

Initial: 0.150 volt maximum.

After life tests: 0.175 volt maximum.

Minimum current: 0.050 ampere.

Contact bounce: See table II.

Overload:

DC: 8 amperes minimum.

AC: Not applicable.

Rupture:

DC: 10 amperes minimum.

AC: Not applicable.

Input data:

Duty ratings: Continuous.

Input voltage:

Maximum: See table II.

Nominal: See table II.

Voltage:

Operate: See table II.

Release: See table II.

Operate time: See table II.

Release time: See table II.

Maximum coil current: See table II.



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TABLE II. Operating characteristics.

| Dash number 1/ | DC coil volts 2/ | | | Maximum dc coil | | Time-milliseconds-maximum | | |
|---|---|---|---|---|---|---|---|---|
| | Nomina | Pickup 3/ | Dropout | Volts | Amps | Operate 4/ | Release delay 5/ | Contact bounce 4/ |
| -001 | 28 | 15.5 +0.5, -0.0 | 14.0 -0.0 6/ | 30 | 0.11 | 10 | 150 ±50 | 1.0 |
| -002 | | 25.75 ±0.25 | 24.75 ±0.25 | | | | 3500 ±500 | |
| -003 | | 21.0 +0.5, -0.0 | 20.5 +0.0, -0.5 | | | | | |
| ⑥ -004 | | 15.5 +0.5, -0.0 | 14.0 -0.0 6/ | | | | 1000 ±-200 | |

1/ The suffix letter W, X, or Y to designate quality level shall be added to the dash number.

2/ Over total operating temperature range.

3/ Applies also to MIL-R-6106 high temperature and continuous current tests.

4/ With nominal coil voltage.

5/ From dropout voltage to 8 volts. Below 8 volts, release delay time may be 0.

6/ Plus tolerance permits any value less than pickup voltage.

Electrical data:

Insulation resistance:

Initial: See table III.

After life: See table III.

Dielectric withstanding voltage (sea level):

Coil to case:

Initial: See table III.

After life: See table III.

All other points:

Initial: See table III.

After life: See table III.

Dielectric withstanding voltage (80,000 feet):

Coil to case:

Initial: See table III.

After life: See table III.



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All other points:

Initial: See table III.

After life: See table III.

Leakage current: See table III.

TABLE III. Electrical data.

| Minimum insulation resistance-megohms | | Dielectric withstanding voltage rms | | | | |
|---|---|---|---|---|---|---|
| | | Sea level 1/ | | | 80,000 feet altitude 2/ | |
| Initial | After tests | | Initial | After tests | Initial | After tests |
| 100 | 100 | Coil to case | 1050 | 1050 | 350 | 350 |
| | | All other points | 600 | 600 | 350 | 350 |
| | | Normal leakage current-rms microamperes 3/ | | | | |
| | | Coil to case | 25 | 25 | | |
| | | All other points | 25 | 25 | | |

1/ Testing performed with relay installed in socket and bracket assembly (M6106/5-005) and secured in mounting track (M6106/3-001).

2/ Relay not installed in its socket and bracket assembly (M6106/5).

3/ Leakage current at sea level.

Transients (applicable to -003 only): The time delay relay shall comply with MIL-STD-461 (applies to X1 and X2 terminals only; A1, A2, and A3 shall be wired but unloaded) and the requirements defined in method CE03, CS01, CS02, CS06, RE02, RS02, and RS03 of MIL-STD-462 except that the field intensity of RS03 shall be in accordance with table IV herein. Voltage spikes (transients of 500 microseconds or less duration) generated by the time delay relay and conducted on lines X1, X2 of the relay shall not exceed ±42 volts. Voltages generated by the time delay relays shall be monitored on the control/power leads (+X1, X2) with an oscilloscope having the ability to detect, measure, and display voltage spikes of 0.25 to 1,000 volts amplitude and 10 nanoseconds to 50 microseconds duration. The oscilloscope shall be capable of retaining the display. The qualification test report shall include photographs of all oscilloscope-displayed spike voltages. Switches used to control the time delay relays shall not be EMI-suppressed in any way. During susceptibility testing, the relay shall not change state nor have the timing interval affected in any way by the EMI signals. The timing interval shall be considered to include the instants of turn-on and turn-off.

TABLE IV. Field intensity levels.

| Frequency | RS03 level (volts/meter) |
|---|---|
| 14 kHz to 200 MHz | 20 |
| 200 MHz to 450 MHz | 65 |
| 450 MHz to 1 GHz | 20 |
| 1 GHz to 12 GHz | 100 |



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Environmental data:

Temperature range (operating): -70°C to +85°C. +171°C nonoperating, 10 minutes. +200°C, nonoperating, 1 minute.

Altitude: 80,000 feet.

Shock:

G level: 50 g's.

Duration: 10 milliseconds.

Contact opening: 10 microseconds, maximum.

Contact closing: 1 microsecond, maximum.

Vibration (sinusoidal):

G level: 30 g's.

Frequency range: 10 - 3,000 Hz.

Vibration (random): Applicable per MIL-STD-810.

Power spector density: 0.4 $g^2$/Hz.

Frequency range: 50 - 2,000 Hz.

Vibration scan:

Frequency range: 5 - 2,000 Hz.

Intensity: 0.01 inch double amplitude or ±2 g's (whichever is less).

Acceleration: 15 g's.

Seal: Applicable to nonhermetically-sealed relays and hermetically-sealed (potted) relays.

Physical data:

Dimensions and configuration: See figure 1.

Weight: 0.24 pound (maximum).

Finish: Orange 32246, FED-STD-595.

Part number: The part number shall be constructed as follows:

| | | | |
|---|---|---|---|
| | M | 83726/27 - 001 | W |
| Military designation | | | |
| Specification sheet number | | | |
| Three digit dash number from table II. | | | |
| Quality level indication (W, X, or Y) 1/ (see MIL-R-83726) | | | |

Example: M83726-001W

1/ Any relays numbered prior to the date of this specification without a quality level indicator shall be considered interchangeable (store and issue) with the "W" quality level.



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## CONCLUDING MATERIAL

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